

TABLE I: Detailed case-level results of MMO_b , MMO_o , and PMO for neural architecture search, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better). Blue denote the best optimizer for each SE case.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
c10mop10_0	1.0	2.0	2.0	0.332 ± 0.178	0.444 ± 0.000	0.499 ± 0.176
c10mop11_0	1.0	1.0	1.0	0.000 ± 0.000	0.100 ± 0.316	0.100 ± 0.316
c10mop12_0	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.653 ± 0.243
c10mop13_0	1.0	2.0	3.0	0.105 ± 0.051	0.148 ± 0.000	0.222 ± 0.285
c10mop1_0	1.0	2.0	3.0	0.419 ± 0.397	0.617 ± 0.212	0.884 ± 0.248
c10mop3_0	1.0	1.0	2.0	0.052 ± 0.163	0.000 ± 0.000	0.668 ± 0.323
c10mop5_0	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.274 ± 0.442
c10mop8_0	1.0	2.0	3.0	0.222 ± 0.151	0.344 ± 0.174	0.511 ± 0.230
citysegmop3_0	1.0	1.0	2.0	0.276 ± 0.152	0.276 ± 0.152	0.559 ± 0.263
in1kmop1_0	1.0	1.0	2.0	0.402 ± 0.205	0.446 ± 0.197	0.633 ± 0.202
in1kmop4_0	1.0	3.0	2.0	0.397 ± 0.320	0.861 ± 0.147	0.723 ± 0.119
in1kmop7_0	1.0	2.0	3.0	0.123 ± 0.090	0.173 ± 0.099	0.554 ± 0.272
Average	1.00	1.67	2.25	0.194 ± 0.142	0.284 ± 0.108	0.523 ± 0.260

TABLE II: Detailed case-level results of MMO_b , MMO_o , and PMO for next release problem. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
nrp-e1	1.0	1.0	2.0	0.016 ± 0.015	0.016 ± 0.015	0.749 ± 0.193
nrp-e2	1.0	1.0	2.0	0.006 ± 0.006	0.006 ± 0.006	0.593 ± 0.163
nrp-e3	1.0	1.0	2.0	0.001 ± 0.002	0.001 ± 0.002	0.489 ± 0.190
nrp-e4	1.0	1.0	2.0	0.005 ± 0.015	0.005 ± 0.015	0.769 ± 0.145
nrp-g1	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.619 ± 0.259
nrp-g2	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.689 ± 0.130
nrp-g3	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.580 ± 0.238
nrp-g4	1.0	1.0	2.0	0.001 ± 0.002	0.001 ± 0.002	0.721 ± 0.168
nrp-m1	1.0	1.0	2.0	0.005 ± 0.007	0.005 ± 0.007	0.779 ± 0.143
nrp-m2	1.0	1.0	2.0	0.010 ± 0.009	0.010 ± 0.009	0.865 ± 0.102
nrp-m3	1.0	1.0	2.0	0.003 ± 0.003	0.003 ± 0.003	0.612 ± 0.176
nrp-m4	1.0	1.0	2.0	0.008 ± 0.006	0.008 ± 0.006	0.762 ± 0.125
nrp1	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.411 ± 0.210
nrp2	1.0	1.0	2.0	0.131 ± 0.118	0.131 ± 0.118	0.634 ± 0.152
nrp3	1.0	1.0	2.0	0.007 ± 0.007	0.007 ± 0.007	0.827 ± 0.086
nrp4	1.0	1.0	2.0	0.001 ± 0.001	0.001 ± 0.001	0.933 ± 0.048
nrp5	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.926 ± 0.049
Average	1.00	1.00	2.00	0.011 ± 0.011	0.011 ± 0.011	0.703 ± 0.152

TABLE III: Detailed case-level results of MMO_b , MMO_o , and PMO for software configuration tuning. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
dnn_adiac	2.0	3.0	1.0	0.471 ± 0.379	0.680 ± 0.353	0.204 ± 0.029
dnn_adiac_reverse	1.0	3.0	2.0	0.026 ± 0.067	0.371 ± 0.348	0.141 ± 0.121
dnn_coffee	2.0	1.0	1.0	0.367 ± 0.477	0.267 ± 0.432	0.000 ± 0.000
dnn_coffee_reverse	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.400 ± 0.516
dnn_dsr	1.0	1.0	2.0	0.102 ± 0.198	0.102 ± 0.198	0.526 ± 0.291
dnn_dsr_reverse	1.0	1.0	1.0	0.000 ± 0.000	0.100 ± 0.316	0.100 ± 0.316
dnn_sa	1.0	3.0	2.0	0.222 ± 0.203	0.565 ± 0.388	0.325 ± 0.071
dnn_sa_reverse	1.0	2.0	3.0	0.000 ± 0.000	0.414 ± 0.356	0.570 ± 0.396
llvm	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
llvm_reverse	1.0	2.0	3.0	0.255 ± 0.158	0.473 ± 0.219	0.691 ± 0.410
lrzip	1.0	1.0	2.0	0.313 ± 0.108	0.303 ± 0.173	0.535 ± 0.313
lrzip_reverse	1.0	1.0	2.0	0.000 ± 0.001	0.000 ± 0.001	0.148 ± 0.306
mariadb	2.0	1.0	1.0	0.298 ± 0.480	0.298 ± 0.480	0.000 ± 0.000
mariadb_reverse	1.0	3.0	2.0	0.046 ± 0.145	0.656 ± 0.311	0.341 ± 0.307
mongodb	1.0	2.0	3.0	0.247 ± 0.319	0.409 ± 0.371	0.556 ± 0.196
mongodb_reverse	1.0	2.0	2.0	0.500 ± 0.527	0.900 ± 0.316	0.900 ± 0.316
storm_rs	1.0	1.0	2.0	0.058 ± 0.184	0.158 ± 0.334	0.825 ± 0.139
storm_rs_reverse	2.0	2.0	1.0	0.011 ± 0.036	0.685 ± 0.403	0.011 ± 0.036
storm_wc	1.0	3.0	2.0	0.000 ± 0.000	0.197 ± 0.347	0.100 ± 0.316
storm_wc_reverse	1.0	3.0	2.0	0.000 ± 0.000	0.625 ± 0.320	0.249 ± 0.350
trimesh	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
trimesh_reverse	1.0	2.0	3.0	0.097 ± 0.051	0.279 ± 0.237	0.568 ± 0.340
vp9	1.0	1.0	2.0	0.055 ± 0.174	0.155 ± 0.343	0.487 ± 0.370
vp9_reverse	1.0	2.0	3.0	0.104 ± 0.139	0.213 ± 0.172	0.472 ± 0.257
x264	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
x264_reverse	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
Average	1.15	1.77	1.85	0.199 ± 0.140	0.379 ± 0.247	0.390 ± 0.207

TABLE IV: Detailed case-level results of MMO_b , MMO_o , and PMO for software effort estimation. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
china_fold1	1.0	3.0	2.0	0.120 ± 0.092	0.373 ± 0.256	0.167 ± 0.071
china_fold2	1.0	3.0	2.0	0.035 ± 0.031	0.540 ± 0.315	0.043 ± 0.033
china_fold3	2.0	2.0	1.0	0.196 ± 0.296	0.199 ± 0.164	0.119 ± 0.066
desharnais_fold1	2.0	1.0	1.0	0.444 ± 0.311	0.444 ± 0.311	0.205 ± 0.162
desharnais_fold2	2.0	1.0	1.0	0.431 ± 0.267	0.431 ± 0.267	0.188 ± 0.169
desharnais_fold3	1.0	1.0	2.0	0.152 ± 0.124	0.152 ± 0.124	0.443 ± 0.270
finnish_fold1	2.0	1.0	1.0	0.517 ± 0.206	0.517 ± 0.206	0.416 ± 0.305
finnish_fold2	1.0	1.0	2.0	0.150 ± 0.105	0.203 ± 0.174	0.288 ± 0.360
finnish_fold3	1.0	3.0	2.0	0.079 ± 0.081	0.342 ± 0.256	0.168 ± 0.101
maxwell_fold1	1.0	1.0	1.0	0.328 ± 0.191	0.328 ± 0.191	0.360 ± 0.303
maxwell_fold2	1.0	3.0	2.0	0.311 ± 0.237	0.738 ± 0.225	0.400 ± 0.142
maxwell_fold3	1.0	3.0	2.0	0.383 ± 0.305	0.578 ± 0.138	0.522 ± 0.172
miyazaki_fold1	2.0	2.0	1.0	0.214 ± 0.162	0.439 ± 0.291	0.229 ± 0.134
miyazaki_fold2	2.0	1.0	1.0	0.423 ± 0.312	0.435 ± 0.206	0.347 ± 0.222
miyazaki_fold3	2.0	3.0	1.0	0.379 ± 0.234	0.447 ± 0.269	0.318 ± 0.268
Average	1.47	1.93	1.47	0.277 ± 0.197	0.411 ± 0.226	0.281 ± 0.185

TABLE V: Detailed case-level results of MMO_b , MMO_o , and PMO for software defect prediction. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
ant-1.7_J48	1.0	2.0	3.0	0.328 ± 0.220	0.485 ± 0.235	0.582 ± 0.207
ant-1.7_KNN	1.0	2.0	2.0	0.418 ± 0.204	0.631 ± 0.219	0.566 ± 0.199
ant-1.7_LR	1.0	1.0	1.0	0.401 ± 0.302	0.429 ± 0.307	0.408 ± 0.304
ant-1.7_NB	1.0	3.0	2.0	0.418 ± 0.228	0.555 ± 0.303	0.468 ± 0.264
camel-1.6_J48	1.0	2.0	3.0	0.353 ± 0.303	0.473 ± 0.308	0.600 ± 0.373
camel-1.6_KNN	1.0	2.0	3.0	0.287 ± 0.139	0.354 ± 0.181	0.571 ± 0.214
camel-1.6_LR	1.0	2.0	2.0	0.419 ± 0.269	0.747 ± 0.237	0.674 ± 0.343
camel-1.6_NB	1.0	2.0	2.0	0.396 ± 0.297	0.556 ± 0.294	0.589 ± 0.314
ivy-2.0_J48	1.0	2.0	3.0	0.549 ± 0.279	0.620 ± 0.304	0.650 ± 0.304
ivy-2.0_KNN	1.0	2.0	2.0	0.446 ± 0.280	0.502 ± 0.293	0.545 ± 0.259
ivy-2.0_LR	1.0	3.0	2.0	0.615 ± 0.269	0.723 ± 0.258	0.680 ± 0.243
ivy-2.0_NB	1.0	2.0	2.0	0.647 ± 0.280	0.687 ± 0.286	0.673 ± 0.286
jedit-4.0_J48	1.0	1.0	2.0	0.496 ± 0.299	0.543 ± 0.342	0.568 ± 0.359
jedit-4.0_KNN	1.0	2.0	2.0	0.428 ± 0.183	0.509 ± 0.190	0.573 ± 0.217
jedit-4.0_LR	1.0	3.0	2.0	0.472 ± 0.301	0.618 ± 0.302	0.575 ± 0.278
jedit-4.0_NB	1.0	2.0	2.0	0.525 ± 0.275	0.588 ± 0.282	0.596 ± 0.302
lucene-2.4_J48	1.0	2.0	3.0	0.416 ± 0.220	0.546 ± 0.245	0.713 ± 0.224
lucene-2.4_KNN	1.0	2.0	3.0	0.472 ± 0.242	0.547 ± 0.209	0.648 ± 0.323
lucene-2.4_LR	1.0	2.0	2.0	0.406 ± 0.269	0.582 ± 0.256	0.537 ± 0.251
lucene-2.4_NB	1.0	2.0	2.0	0.401 ± 0.236	0.492 ± 0.254	0.465 ± 0.255
poi-3.0_J48	1.0	2.0	3.0	0.394 ± 0.268	0.456 ± 0.245	0.546 ± 0.314
poi-3.0_KNN	1.0	2.0	2.0	0.641 ± 0.268	0.733 ± 0.290	0.729 ± 0.285
poi-3.0_LR	1.0	3.0	2.0	0.636 ± 0.269	0.781 ± 0.208	0.682 ± 0.253
poi-3.0_NB	1.0	2.0	2.0	0.576 ± 0.254	0.674 ± 0.214	0.623 ± 0.241
synapse-1.2_J48	1.0	2.0	2.0	0.374 ± 0.264	0.528 ± 0.286	0.586 ± 0.291
synapse-1.2_KNN	1.0	2.0	3.0	0.366 ± 0.283	0.450 ± 0.228	0.493 ± 0.245
synapse-1.2_LR	1.0	2.0	2.0	0.466 ± 0.255	0.557 ± 0.253	0.547 ± 0.302
synapse-1.2_NB	1.0	2.0	2.0	0.494 ± 0.294	0.546 ± 0.284	0.542 ± 0.274
velocity-1.6_J48	1.0	2.0	3.0	0.279 ± 0.179	0.411 ± 0.176	0.584 ± 0.230
velocity-1.6_KNN	1.0	2.0	2.0	0.409 ± 0.243	0.510 ± 0.301	0.572 ± 0.296
velocity-1.6_LR	1.0	2.0	2.0	0.361 ± 0.288	0.510 ± 0.310	0.484 ± 0.288
velocity-1.6_NB	1.0	1.0	1.0	0.449 ± 0.302	0.496 ± 0.296	0.501 ± 0.302
xalan-2.6_J48	1.0	2.0	3.0	0.410 ± 0.292	0.535 ± 0.302	0.604 ± 0.314
xalan-2.6_KNN	1.0	2.0	3.0	0.252 ± 0.231	0.507 ± 0.222	0.584 ± 0.269
xalan-2.6_LR	1.0	3.0	2.0	0.472 ± 0.297	0.669 ± 0.295	0.535 ± 0.279
xalan-2.6_NB	1.0	2.0	2.0	0.460 ± 0.312	0.572 ± 0.339	0.525 ± 0.323
xerces-1.4_J48	1.0	1.0	1.0	0.536 ± 0.319	0.581 ± 0.344	0.587 ± 0.352
xerces-1.4_KNN	1.0	3.0	2.0	0.320 ± 0.305	0.450 ± 0.286	0.384 ± 0.276
xerces-1.4_LR	1.0	3.0	2.0	0.398 ± 0.262	0.527 ± 0.263	0.456 ± 0.250
xerces-1.4_NB	1.0	1.0	1.0	0.527 ± 0.317	0.529 ± 0.318	0.527 ± 0.317
Average	1.00	2.05	2.17	0.443 ± 0.265	0.555 ± 0.269	0.569 ± 0.280

TABLE VI: Detailed case-level results of MMO_b , MMO_o , and PMO for software product lines testing. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
7z	1.0	1.0	2.0	0.133 ± 0.070	0.133 ± 0.070	0.517 ± 0.228
amazon	1.0	1.0	2.0	0.308 ± 0.159	0.308 ± 0.159	0.719 ± 0.195
berkeleydbc	1.0	1.0	2.0	0.229 ± 0.074	0.229 ± 0.120	0.514 ± 0.254
cocheecologico	1.0	1.0	2.0	0.229 ± 0.139	0.229 ± 0.139	0.765 ± 0.150
counterstrikesimplefeaturemodel	1.0	1.0	2.0	0.164 ± 0.158	0.171 ± 0.108	0.579 ± 0.195
drupal	1.0	2.0	3.0	0.338 ± 0.176	0.370 ± 0.177	0.787 ± 0.141
dssample	1.0	1.0	2.0	0.167 ± 0.159	0.167 ± 0.159	0.633 ± 0.235
dune	1.0	2.0	3.0	0.300 ± 0.169	0.425 ± 0.134	0.713 ± 0.145
electronicdrum	1.0	1.0	2.0	0.280 ± 0.131	0.280 ± 0.131	0.764 ± 0.183
hipacc	1.0	1.0	2.0	0.243 ± 0.141	0.280 ± 0.120	0.630 ± 0.292
javagc	1.0	2.0	3.0	0.100 ± 0.096	0.243 ± 0.142	0.591 ± 0.194
jhipster	1.0	1.0	2.0	0.245 ± 0.157	0.245 ± 0.157	0.561 ± 0.228
lrzip	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
modeltransformation	1.0	1.0	2.0	0.250 ± 0.172	0.295 ± 0.099	0.645 ± 0.223
polly	1.0	2.0	3.0	0.305 ± 0.150	0.365 ± 0.163	0.755 ± 0.185
smarthomev2.2	1.0	2.0	3.0	0.196 ± 0.193	0.284 ± 0.143	0.761 ± 0.144
splssimuelespnp	1.0	2.0	3.0	0.175 ± 0.101	0.231 ± 0.072	0.613 ± 0.266
videoplayer	1.0	1.0	2.0	0.263 ± 0.163	0.263 ± 0.163	0.838 ± 0.120
vp9	1.0	2.0	3.0	0.312 ± 0.137	0.428 ± 0.107	0.740 ± 0.146
webportal	1.0	1.0	2.0	0.125 ± 0.101	0.125 ± 0.101	0.547 ± 0.282
x264	1.0	2.0	3.0	0.225 ± 0.184	0.275 ± 0.142	0.575 ± 0.237
Average	1.00	1.38	2.33	0.242 ± 0.135	0.278 ± 0.124	0.655 ± 0.192

TABLE VII: Detailed case-level results of MMO_b , MMO_o , and PMO for software project scheduling. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
10-10-skill-4-5	1.0	1.0	2.0	0.012 ± 0.007	0.012 ± 0.007	0.841 ± 0.110
10-10-skill-6-7	1.0	1.0	2.0	0.005 ± 0.005	0.005 ± 0.005	0.748 ± 0.137
10-15-skill-4-5	1.0	1.0	2.0	0.009 ± 0.010	0.005 ± 0.003	0.907 ± 0.064
10-15-skill-6-7	1.0	2.0	3.0	0.003 ± 0.003	0.004 ± 0.003	0.904 ± 0.065
10-5-skill-4-5	1.0	1.0	2.0	0.001 ± 0.002	0.001 ± 0.002	0.772 ± 0.113
10-5-skill-6-7	1.0	2.0	3.0	0.008 ± 0.007	0.023 ± 0.044	0.612 ± 0.162
20-10-skill-4-5	1.0	1.0	2.0	0.011 ± 0.008	0.009 ± 0.008	0.842 ± 0.067
20-10-skill-6-7	1.0	1.0	2.0	0.007 ± 0.005	0.005 ± 0.004	0.853 ± 0.085
20-15-skill-4-5	1.0	1.0	2.0	0.014 ± 0.008	0.014 ± 0.008	0.872 ± 0.065
20-15-skill-6-7	1.0	1.0	2.0	0.030 ± 0.020	0.032 ± 0.015	0.940 ± 0.048
20-5-skill-4-5	1.0	1.0	2.0	0.022 ± 0.016	0.022 ± 0.016	0.807 ± 0.120
20-5-skill-6-7	1.0	2.0	3.0	0.010 ± 0.007	0.012 ± 0.004	0.825 ± 0.124
30-10-skill-4-5	1.0	1.0	2.0	0.012 ± 0.009	0.011 ± 0.007	0.855 ± 0.075
30-10-skill-6-7	1.0	1.0	2.0	0.010 ± 0.007	0.010 ± 0.007	0.880 ± 0.068
30-15-skill-4-5	1.0	1.0	2.0	0.018 ± 0.015	0.020 ± 0.015	0.874 ± 0.096
30-15-skill-6-7	1.0	2.0	3.0	0.016 ± 0.014	0.020 ± 0.013	0.901 ± 0.057
30-5-skill-4-5	1.0	2.0	3.0	0.030 ± 0.019	0.036 ± 0.016	0.831 ± 0.091
30-5-skill-6-7	1.0	2.0	3.0	0.014 ± 0.022	0.021 ± 0.018	0.874 ± 0.110
Average	1.00	1.33	2.33	0.013 ± 0.010	0.015 ± 0.011	0.841 ± 0.092

TABLE VIII: Detailed case-level results of MMO_b , MMO_o , and PMO for third party library migration. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
migrationRule10_ALL	2.0	3.0	1.0	0.100 ± 0.316	0.202 ± 0.261	0.000 ± 0.000
migrationRule10_CO	1.0	1.0	1.0	0.100 ± 0.316	0.100 ± 0.316	0.100 ± 0.316
migrationRule10_DS	2.0	3.0	1.0	0.012 ± 0.038	0.224 ± 0.412	0.000 ± 0.000
migrationRule10_MS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_ALL	1.0	1.0	2.0	0.032 ± 0.102	0.064 ± 0.136	0.404 ± 0.264
migrationRule18_CO	1.0	1.0	1.0	0.100 ± 0.316	0.000 ± 0.000	0.000 ± 0.000
migrationRule18_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_MS	1.0	1.0	2.0	0.017 ± 0.053	0.017 ± 0.053	0.217 ± 0.284
migrationRule1_ALL	1.0	1.0	2.0	0.115 ± 0.064	0.136 ± 0.153	0.663 ± 0.229
migrationRule1_CO	1.0	2.0	3.0	0.085 ± 0.135	0.113 ± 0.146	0.623 ± 0.282
migrationRule1_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule1_MS	1.0	1.0	2.0	0.040 ± 0.126	0.080 ± 0.169	0.760 ± 0.207
migrationRule2_ALL	1.0	2.0	3.0	0.076 ± 0.050	0.121 ± 0.117	0.881 ± 0.093
migrationRule2_CO	1.0	1.0	2.0	0.151 ± 0.128	0.151 ± 0.128	0.793 ± 0.135
migrationRule2_DS	1.0	2.0	3.0	0.126 ± 0.062	0.143 ± 0.091	0.850 ± 0.087
migrationRule2_MS	1.0	1.0	2.0	0.104 ± 0.056	0.104 ± 0.056	0.804 ± 0.139
migrationRule3_ALL	1.0	2.0	3.0	0.282 ± 0.303	0.441 ± 0.141	0.698 ± 0.222
migrationRule3_CO	1.0	2.0	2.0	0.485 ± 0.337	0.712 ± 0.139	0.693 ± 0.131
migrationRule3_MS	1.0	2.0	3.0	0.125 ± 0.177	0.225 ± 0.184	0.475 ± 0.275
migrationRule4_ALL	1.0	1.0	2.0	0.096 ± 0.092	0.120 ± 0.096	0.704 ± 0.173
migrationRule4_CO	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
migrationRule4_DS	1.0	1.0	1.0	0.100 ± 0.316	0.000 ± 0.000	0.000 ± 0.000
migrationRule4_MS	1.0	1.0	2.0	0.000 ± 0.000	0.067 ± 0.211	0.600 ± 0.377
migrationRule5_ALL	2.0	2.0	1.0	0.260 ± 0.324	0.348 ± 0.292	0.203 ± 0.133
migrationRule5_CO	1.0	1.0	1.0	0.350 ± 0.337	0.350 ± 0.337	0.400 ± 0.211
migrationRule5_DS	1.0	1.0	2.0	0.100 ± 0.211	0.259 ± 0.280	0.195 ± 0.345
migrationRule5_MS	1.0	1.0	1.0	0.100 ± 0.316	0.100 ± 0.316	0.100 ± 0.316
migrationRule7_ALL	1.0	1.0	2.0	0.228 ± 0.094	0.282 ± 0.182	0.849 ± 0.144
migrationRule7_CO	1.0	1.0	2.0	0.229 ± 0.162	0.240 ± 0.134	0.852 ± 0.127
migrationRule7_DS	1.0	2.0	3.0	0.215 ± 0.109	0.258 ± 0.069	0.826 ± 0.151
migrationRule7_MS	1.0	1.0	2.0	0.164 ± 0.118	0.164 ± 0.118	0.764 ± 0.148
migrationRule8_ALL	1.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.617 ± 0.209
migrationRule8_CO	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.200 ± 0.422
migrationRule8_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule8_MS	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.400 ± 0.516
Average	1.09	1.40	1.83	0.165 ± 0.133	0.201 ± 0.130	0.505 ± 0.170

TABLE IX: Detailed case-level results of MMO_b , MMO_o , and PMO for workflow scheduling. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
CyberShake_100	1.0	2.0	3.0	0.017 ± 0.012	0.055 ± 0.029	0.442 ± 0.269
CyberShake_1000	1.0	2.0	3.0	0.186 ± 0.148	0.291 ± 0.120	0.779 ± 0.201
CyberShake_30	1.0	2.0	3.0	0.017 ± 0.017	0.046 ± 0.051	0.483 ± 0.276
CyberShake_50	1.0	2.0	3.0	0.045 ± 0.032	0.096 ± 0.033	0.436 ± 0.299
Epigenomics_100	1.0	1.0	2.0	0.195 ± 0.176	0.195 ± 0.176	0.270 ± 0.274
Epigenomics_24	1.0	2.0	3.0	0.014 ± 0.006	0.022 ± 0.006	0.152 ± 0.309
Epigenomics_46	1.0	2.0	3.0	0.112 ± 0.120	0.249 ± 0.199	0.631 ± 0.246
Epigenomics_997	1.0	1.0	2.0	0.328 ± 0.211	0.308 ± 0.224	0.738 ± 0.193
Inspirai_100	1.0	2.0	3.0	0.070 ± 0.063	0.098 ± 0.054	0.653 ± 0.238
Inspirai_1000	1.0	1.0	2.0	0.304 ± 0.163	0.304 ± 0.163	0.719 ± 0.197
Inspirai_30	1.0	2.0	3.0	0.000 ± 0.000	0.000 ± 0.000	0.512 ± 0.248
Inspirai_50	1.0	2.0	3.0	0.052 ± 0.060	0.078 ± 0.046	0.687 ± 0.216
Montage_100	1.0	2.0	3.0	0.052 ± 0.037	0.081 ± 0.039	0.762 ± 0.203
Montage_1000	1.0	1.0	2.0	0.295 ± 0.129	0.351 ± 0.295	0.828 ± 0.112
Montage_25	1.0	2.0	3.0	0.044 ± 0.100	0.065 ± 0.023	0.556 ± 0.277
Montage_50	1.0	2.0	3.0	0.030 ± 0.023	0.055 ± 0.031	0.391 ± 0.247
Sipht_100	1.0	1.0	2.0	0.003 ± 0.003	0.003 ± 0.003	0.294 ± 0.326
Sipht_1000	1.0	2.0	3.0	0.140 ± 0.104	0.225 ± 0.134	0.680 ± 0.198
Sipht_30	1.0	2.0	3.0	0.008 ± 0.005	0.018 ± 0.012	0.150 ± 0.299
Sipht_60	1.0	2.0	3.0	0.006 ± 0.004	0.064 ± 0.073	0.175 ± 0.298
Average	1.00	1.75	2.75	0.096 ± 0.071	0.130 ± 0.086	0.517 ± 0.246

TABLE X: Detailed case-level results of MMO_b , MMO_o , and PMO for web service composition. The formate is the same as Table I.

Dataset	Rank			Mean $\pm$ Std		
	MMO_b	MMO_o	PMO	MMO_b	MMO_o	PMO
workflow_1	1.0	1.0	2.0	0.150 ± 0.112	0.150 ± 0.112	0.659 ± 0.168
workflow_10	1.0	1.0	2.0	0.275 ± 0.183	0.284 ± 0.161	0.526 ± 0.183
workflow_10_reverse	1.0	2.0	3.0	0.374 ± 0.202	0.426 ± 0.115	0.622 ± 0.322
workflow_1_reverse	1.0	2.0	3.0	0.479 ± 0.214	0.576 ± 0.242	0.648 ± 0.169
workflow_2	1.0	1.0	2.0	0.155 ± 0.135	0.155 ± 0.135	0.638 ± 0.261
workflow_2_reverse	1.0	2.0	3.0	0.372 ± 0.198	0.543 ± 0.195	0.716 ± 0.165
workflow_3	1.0	2.0	3.0	0.251 ± 0.203	0.401 ± 0.113	0.704 ± 0.206
workflow_3_reverse	1.0	2.0	3.0	0.307 ± 0.183	0.560 ± 0.208	0.627 ± 0.269
workflow_4	1.0	1.0	2.0	0.274 ± 0.219	0.250 ± 0.195	0.616 ± 0.286
workflow_4_reverse	1.0	1.0	2.0	0.441 ± 0.192	0.441 ± 0.192	0.701 ± 0.187
workflow_5	1.0	2.0	3.0	0.220 ± 0.155	0.335 ± 0.175	0.743 ± 0.163
workflow_5_reverse	1.0	2.0	2.0	0.440 ± 0.267	0.598 ± 0.256	0.641 ± 0.158
workflow_6	1.0	1.0	2.0	0.166 ± 0.168	0.166 ± 0.168	0.625 ± 0.235
workflow_6_reverse	1.0	1.0	2.0	0.498 ± 0.237	0.538 ± 0.209	0.723 ± 0.182
workflow_7	1.0	1.0	2.0	0.347 ± 0.185	0.347 ± 0.181	0.725 ± 0.205
workflow_7_reverse	1.0	1.0	2.0	0.587 ± 0.239	0.652 ± 0.199	0.796 ± 0.146
workflow_8	1.0	1.0	2.0	0.230 ± 0.146	0.230 ± 0.146	0.547 ± 0.232
workflow_8_reverse	1.0	2.0	3.0	0.466 ± 0.240	0.590 ± 0.109	0.738 ± 0.153
workflow_9	1.0	2.0	3.0	0.261 ± 0.165	0.354 ± 0.149	0.703 ± 0.242
workflow_9_reverse	1.0	3.0	2.0	0.442 ± 0.239	0.651 ± 0.204	0.603 ± 0.247
Average	1.00	1.55	2.40	0.337 ± 0.194	0.412 ± 0.173	0.665 ± 0.209